US Technology

AI Data Center Reality Check – key takeaways from expert call

Equities
Software

United States

- Demand is strong, but project deliveries can slip
- Power, people, and permits are creating constraints
- Behind-the-meter generation has become the default

We hosted an expert call with Randy Dentinger, Vice President of Procurement & Supply Chain at Takanock, LLC on 15 October 2026.

The deals in the pipeline can slip, but the demand signals are not. Mr Dentinger distinguishes between the headline and deliverable pipelines. He estimates that 40-50% of proposed projects are delayed, with cancellations and delays running at 3-4x last year's level, but around half of the backlog is speculative builds. The interconnection queue provides a precedent, with 72% of grid connection requests historically never materializing, yet vacancy is c1% nationally, tighter in Northern Virginia, and 84% of capacity is pre-leased before completion. In Abilene, OpenAI and *Oracle* passed on c600MWs, however, the project was re-signed with another hyperscaler within days at a higher rate. Cancellations mostly reflect power or permitting problems, not weak demand. Even at a 50% completion rate, the current pipeline would support a substantial buildout.

Power first, then people, then permits. Five-to-seven-year grid queues and repeatedly extended utility quotes are making behind-the-meter generation standard, either as bridging power or as a permanent islanded supply. Gas turbines, aeroderivative engines, and reciprocating engines are on record order books at *GE Vernova*, *Siemens Energy*, *Caterpillar*, and *Baker Hughes*, with availability extending towards 2032. People are the next constraint, with limited skilled labor slowing project delivery. Permits rank third, but they remain a material source of delay, particularly where projects require complex environmental, land-use, air-quality, and local approvals (see *Beyond power: AI and data centre sustainability*, 12 Oct). Equipment adds further pressure, with turbines, transformers, switchgear, and circuit breakers on extended lead times, while turbine availability is reaching towards 2032.

Geography, cost, and the technologies to watch. Developers are targeting flared gas, unconnected wind, and emerging metros, while tier 1 cities still need inference capacity close to users. Persistent US constraints could push capacity towards Canada, northern Mexico, Scandinavia, the Middle East, and Asia. The next nine largest countries combined now roughly equal US capacity. Commodity inflation, record copper prices, diesel at USD6.50 per gallon, and utility bills up 6-7% nationally, including 20-22% in New Jersey, are increasing political pressure. Take-or-pay terms and curtailment rights are adding to on-site backup requirements. Technologies to watch are small modular reactors, fuel cells, packaged power modules, combined-cycle conversions, and NVIDIA's 800VDC architecture, although the supply chain still needs solid-state transformers.

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Name	Ticker	Last Close	Rating	TP	Currency
Baker Hughes	BKR US	55.70	Buy	85.00	USD
Caterpillar	CAT US	730.76	Buy	1,150.00	USD
GE Vernova	GEV US	921.20	Hold	1,040.00	USD
NVIDIA	NVDA US	214.13	Buy	365.00	USD
Oracle	ORCL US	128.23	Buy	316.00	USD
Siemens Energy	ENR GR	133.14	Buy	205.00	EUR
SpaceX	SPCX US	146.03	Hold	150.00	USD
American Supercritical	Not Public	NA	NA	NA	NA
APR Energy	Not Public	NA	NA	NA	NA
Crusoe	Not Public	NA	NA	NA	NA
Ecogen Power Systems	Not Public	NA	NA	NA	NA
Elsewedy Electric	SWDY EG	105.55	NA	NA	EGP
OpenAI	Not Public	NA	NA	NA	NA
Regal Rexnord	RRX US	154.01	NA	NA	USD
Thomson Power Systems	Not Public	NA	NA	NA	NA

Source: LSEG Datastream, HSBC, Priced on 16 October 2026

The investor lens

The central debate was whether the data-center opportunity is facing a genuine demand problem or simply a delivery problem. Mr Dentinger's view was that demand remains exceptionally strong, but the "headline pipeline", which includes speculative deals, is not deliverable in full. Around 40-50% of proposed projects could be delayed, however, roughly half of the capacity on the books may be speculative, and delays and cancellations are running at three to four times last year's level. The counterpoint is that vacancy remains c1% nationally, with 84% of capacity pre-leased before completion. Notably, projects that do reach permitting and construction rarely struggle to find a tenant. Therefore, the debate centers on what ultimately constrains growth; demand, power, permitting, equipment or labor, and whether behind-the-meter generation is a temporary bridge or a permanent feature of the industry. Geography is also becoming more important, as US constraints could push capacity towards Canada, Mexico, Scandinavia, the Middle East, and Asia.

For long-only investors, the key message is that the secular datacenter and AI infrastructure growth story remains intact, but the path will be uneven and increasingly shaped by execution. Headline project cancellations may be tied to speculative projects, so knowing the details of the deal is important. Mr Dentinger thinks data center demand could continue to grow at 60-80% through 2027, with slower but still positive growth in 2028 and 2029. However, five-to-seven-year grid queues, transformer and switchgear shortages, skilled-labor constraints, complex permitting, and turbine availability extending towards 2032, will all determine which projects see daylight. The most attractive exposure is likely to be among suppliers of scarce equipment with credible manufacturing capacity and limited China content, including gas turbines, transformers, switchgear, generators, fuel cells, and packaged-power systems. Longer-term opportunities include small modular reactors, fuel cells, combined-cycle conversions and NVIDIA's 800VDC architecture, but the near-term investment case remains centered on securing power and delivering capacity.

For hedge funds, the opportunity is driven by catalysts, relative performance and changes in expectations rather than simply the direction of overall demand. The most important near-term indicators are hyperscaler capex, vacancy rates, pre-leasing, project cancellations, permitting outcomes, utility terms, and equipment lead times. Neoclouds and leased capacity appear to be the first potential pressure point if the market softens, as leases may be easier to exit than developers suggest. By contrast, we think that permitted projects with secured power have a high probability of completion. Potential dislocations include beneficiaries of prolonged turbine, transformer, and switchgear scarcity vs developers exposed to speculative projects or rising

utility costs. Investors should also monitor regional policy changes, China-content restrictions and offshore migration, while using events such as the Abilene re-leasing of roughly 600MW at a higher rate as evidence that an apparent customer withdrawal may represent a shift in location or counterparty, not destruction of demand.

Key takeaways at a glance

Pipeline, delays and cancellations

- c40-50% of proposed projects get delayed, a higher share than get cancelled. Delays and cancellations are running at three to four times last year's pace.
- c50% of the book are speculative. Historically, 72% of grid interconnection requests never come to fruition, and he sees no reason why this cohort would behave differently.
- Growth slows but does not stop. Data center growth has run at 60%-80%. He expects that to hold through 2027 and decelerate into 2028-2029, while still growing.
- Permitting is the main kill mechanism, particularly air permits and local approvals. Developers now run community campaigns “almost like an election”.

Demand indicators remain tight

- c1% national vacancy, lower in Northern Virginia. 84% of capacity is pre-leased before completion, increasingly on land and power alone.
- Nobody walks away late. He has seen no developer abandon a permitted project for lack of a tenant. Walk-aways happen only where permits look unobtainable.
- Abilene is the tell. OpenAI and *Oracle* declined c600MW of incremental capacity and the developer re-signed another hyperscaler within days at a higher rate.
- Watch the neoclouds first. If a slowdown comes, it will likely show up where hyperscalers lease rather than own as those leases are easier to exit than generally assumed.

Power is the binding constraint

- Grid queues of five-to-seven years, with four-year quotes often extended, have made behind-the-meter gas the default bridge and sometimes the permanent solution.
- Record turbine order books at *GE Vernova*, *Siemens Energy*, *Caterpillar*, and *Baker Hughes*, according to company reports. Availability quoted as far out as 2032.
- The supply response is structurally slow. A new US plant would not be built and staffed before 2030, and OEMs lack visibility on hyperscaler capex beyond that horizon.
- Utilities are rewriting terms. Take-or-pay agreements and curtailment rights, typically capped at c20-25% of capacity and c1% of hours, increases the need for more backup generation onto the site.
- Gas is recovering off a low base. Approximately 80% of new US generation over the last five years was renewable. Gas fell to c3GW and may reach c10GW this year against c40GW of solar.

Supply chain and sourcing

- Lead times have extended across the board: switchgear, gas turbines, reciprocating engines, generators, packaged power modules, transformers, GIS switchgear, and circuit breakers.
- Executive Order 14421 and FEOC rules are a second squeeze, removing Chinese content, firmware, and controllers from grid-tied equipment, and creating a rush onto unaffected suppliers.
- China dominates the base. Nine times the structural steel of the next largest producer, and the leading position in inverters, batteries, and renewables equipment.
- Sourcing is shifting rather than reshoring outright. Mexico is now the top US trading partner. India, Europe, South America, Korea, Taiwan, and Southeast Asia are growing, with Elsewedy of Egypt an emerging supplier.
- US capacity is coming, slowly. Siemens Energy's Charlotte transformer plant runs behind schedule with four years of backlog before it is fully operational.

Labor

- People rank ahead of permits. Medium-voltage electricians are the scarcest trade, and errors risk destroyed equipment, injury, or fatality.
- Demographics are against it. Approximately 11,000 Americans retire weekly, c34% of construction labor has historically been foreign-born, and only c5% of high-school students are steered toward the trades.
- Pay is responding. Travelling electricians can earn cUSD250k and skilled mechanical trades cUSD150k. He sees a route here to rebuilding a middle class without four years of college debt.
- Manufacturing labor is the hidden constraint. Equipment makers in rural areas have land and backlog but cannot staff expansion.

Geography and cost

- Siting follows stranded power: flared West Texas gas, unconnected wind farms, and pockets of grid capacity around emerging metros. Tier 1 cities still need low-latency inference capacity.
- Constrained markets. New York's one-year moratorium with sub-50MW still permitted, California's long-standing 100MW limit, and Pennsylvania legislation that Mr Dentinger reads as stricter than the Texas ERCOT audit.
- Offshore migration is a live risk: Canada, Mexico, Scandinavia, the Middle East, and Asian AI corridors. The next nine largest countries combined roughly equal US capacity.
- Cost inflation is broad: record copper prices, higher aluminium and steel prices, diesel at USD6.50/gallon, and utility bills up 6-7% nationally and 20-22% in New Jersey.

Technologies to watch

- Small modular reactors, newly credible on hyperscaler funding, and net-zero shortfalls. For context, the US built two nuclear plants in 30 years against 10-11 Chinese approvals a year.
- Fuel cells and packaged power modules, both capacity-constrained. New entrants are winning large orders, including a USD735m award to Regal Rexnord's Thomson Power Systems.
- Simple-to-combined-cycle conversion as the next efficiency lever if the power crunch tightens further.
- NVIDIA is driving pilots of next-generation 800 VDC power architectures for its Vera Rubin chips in early 2027. The architecture could reduce copper usage by 40-45% and lower conversion losses, although the supply chain, particularly for solid-state transformers, is not yet ready for anticipated demand.

How much of the pipeline actually slips

No precise number, but delays dominate cancellations. Mr Dentinger was clear that the proposed totals are not deliverable. Data center pipeline capacity is staggering, but there is no realistic path to building all of it. His working estimate is that c40-50% will be delayed, with cancellations a smaller subset.

The rate of change is the news. Cancellations and delays this year are running at 3x-4x last year's level, and in his view are snowballing. Legislative changes in Texas, Pennsylvania, and New York are contributing, and large AI projects already under construction are struggling to secure permits.

NIMBY opposition has never been higher. The objections cluster around water, energy, noise, and vibration, with what he regards as a good deal of misinformation attached. Mr Dentinger's conclusion for operators is structural. Hyperscalers can no longer buy land quietly and stay out of view. Transparency about what is being built, and about what the community gets from it, is now a condition of delivery.

What actually kills or delays a project

Permitting, and air permitting in particular. The decisive variable is local approval. Developers must now go into communities and run what amounts to an election campaign, understanding what the community wants and how to earn local support.

Legislative risk has been added on top. Mr Dentinger mentioned the ERCOT audit in Texas as something few expected, noting the key political trigger. The governor's position shifted markedly, only after Willie Nelson came out against data centers. He reads at least part of this as pre-midterm posturing that may prove temporary. Nonetheless, the rhetoric is real and so are the resulting delays.

The community economics, and the jobs argument. Incentives have historically pulled projects in. Mr Dentinger leaned on Abilene, Texas as his example: a city of c130,000 hosting the 1.8GW Stargate flagship for OpenAI and *Oracle*, with 8,000-10,000 construction workers on site. Housing is full and hotels and restaurants are booming. His point is that the incentive was competitive. If Abilene had not offered it, somewhere else would have.

The "no long-term jobs" critique is wrong at current scale. What is under roof at Abilene is larger than 25 superstores on c1,200 acres, with hundreds of transformers, switchgear, and generation equipment on site. All of it has to be maintained to five-nines standards, which supports permanent operations and maintenance teams plus an ecosystem of refurbishment, repair, and inspection contractors. Add behind-the-meter rotating equipment such as gas turbines and reciprocating engines, plus GPU refresh cycles, and the site generates continuing employment rather than a build-and-leave profile.

Which markets carry the most risk

Constrained by legislation. New York is operating a one year moratorium, though sub-50MW facilities can still be built, and California has had an effective 100MW ceiling for some time. The Northeast and West Coast are difficult generally, and opposition to fossil fuels makes behind-the-meter natural gas largely unavailable as a workaround. Mr Dentinger reads Pennsylvania's legislation as somewhat stricter than the Texas equivalent.

Tier-one cities still need capacity regardless. Dallas-Fort Worth, Northern Virginia, Phoenix, Columbus, Atlanta, and Chicago are short of land and power but still require data centers, and specifically inference capacity in and around the metro, which becomes more important as the models improve and monetization scales.

How much of the book is speculative? c50%

Mr Dentinger estimates roughly half of it. Measured by gigawatts rather than project count, he says c50% of the capacity currently on the books lacks a firm commitment behind it. Developers are casting wide nets and pursuing multiple sites because they recognize that not every project will secure permits, obtain power, or ultimately make economic sense.

The interconnection queue is the precedent. Connection timelines now run to five, six, or seven years in some locations, partly because the queue itself is congested with requests. Historically, 72% of grid connection requests have never come to fruition, which makes a 50% non-completion rate for speculative data center proposals a conservative read.

Even half is a large number. Growth has run at 60-80% in recent years. He expects 2027 to stay in that band, with the percentage rate decelerating into 2028 and 2029, while remaining positive. The economics simply stop penciling for some developers when they are quoted six or seven years to connect, and hyperscalers will not wait that long.

When a project becomes impossible to abandon

Almost nobody walks away late. With c1% vacancy and universal demand, a developer holding permits and approvals on a project already underway would never abandon it. The one circumstance in which he sees walk-aways is regulatory, where there are no permits and no credible path to them.

Tenant risk is not the issue. He has not seen anyone walk away because leasing looked difficult. The leading indicators run the other way, with 84% of data centers pre-leased before completion and c1% of capacity available nationally.

Where opposition has bitten. He is watching Project Jupiter in Doña Ana County, New Mexico, as the test case of a mature project that could still stop. It is currently delayed over air and gas permits, even after switching from reciprocating gas engines to cleaner fuel cells. The more common pattern is sequencing. A large project gets approved and built in a specified location, after which the community organizes and declares itself closed to further development. The opposition facing the next entrant is materially stronger than the initial. Developers respond by casting wide nets, developing two sites while expecting one to proceed.

Equipment lead times and the China content squeeze

Lead times have extended across every critical category. Suppliers of switchgear, tier-one OEM gas turbines, reciprocating gas engines for behind-the-meter power, generators and customized power modules are increasingly booked out. Transformers, GIS switchgear, and circuit breakers have seen particularly sharp extensions in recent months.

Executive Order 14421 is a compounding factor. It removes the option to source grid-tied electrical equipment containing Chinese components, firmware, controllers, or software. This not only affects battery energy systems and inverters, but also transformers and breakers. In practice, the order has driven a surge of orders and enquiries towards manufacturers without Chinese exposure, heightening the effective bottlenecks around these components.

The starting point is Chinese dominance. China is the leading manufacturer across this equipment set, particularly inverters, batteries, and anything related to solar and renewables. He offered structural steel as the scale marker, at nine times the output of the second largest producer, alongside electric vehicles. Restricting that supply unsurprisingly creates market-wide concern about availability.

Where the equipment now comes from

Not enough of it is made domestically. Historically c75% of large main power transformers have been imported, and the imported share of smaller medium-voltage pad-mounted transformers had risen to c50%. That trend is now reversing.

China-plus rather than pure reshoring. Mexico is now the top US trading partner. Sourcing is broadening into India, Europe, South America, Korea, Taiwan, and Southeast Asia, with Elsewedy Electric of Egypt emerging as a significant supplier. Chinese manufacturers have in parallel moved assembly into Malaysia, Cambodia, Thailand, and Vietnam, and increasingly Africa, to route under free trade agreements. Some are opening in Mexico and the US, and in certain cases are selling down Chinese ownership toward the 25% threshold that would satisfy covered-foreign-entity restrictions.

Domestic capacity is arriving late. Siemens Energy's new transformer factory in Charlotte is behind schedule and already carries four years of backlog before it is fully up and running.

The tax angle. Projects that began before 4 July 2026 remain eligible for Inflation Reduction Act investment and production tax credits. After that date, full compliance with the new restrictions is required, with FEOC among the most significant.

Power, people, and permits: what resolves and what does not

Power, people, and permits, in that order. Power is the dominant problem. The objective is a grid connection with five-nines reliability, but being quoted five to seven years is not a workable solution.

Hence behind-the-meter power as the default. Whether as bridging supply until grid connection or as the permanent solution, this has produced record order volumes at *GE Vernova*, *Siemens Energy*, *Caterpillar*, and *Baker Hughes* for gas turbines, aeroderivatives, and large reciprocating engines. This is not equipment that can be sourced from Chinese fabricators, and it is not quickly replaced.

Why the OEMs are not racing to add capacity. Many shut gas turbine capacity around 2022, when little fossil generation was being built. Roughly 80% of new US generation over the last five years has been renewable, with gas falling to c3GW against c40GW of solar this year, and gas only now recovering toward c10GW. Rebuilding takes until 2030 or beyond once labor is factored in, at which point hyperscaler capex visibility is low. Today, capex is at record levels and next year's could approach USD1trn, but there is little clarity past 2030, and the OEMs fear committing to large plant expansions into a backdrop of declining volume. Mr Dentinger also flagged a shortage of EPCs able to build large gas projects.

The race framing. Everyone is sprinting to win the AI war, and dominating capacity means dominating power. He cited SpaceX's publicly announced USD3bn for power generation equipment at IPO, a figure many consider far too low, alongside acquisitions such as APR Energy and purchases of turbines, reciprocating engines, and fuel cells. The requirement is not only compute but gas compression for a weekly launch cadence. He expects more than one winner, but the picture becomes opaque in the 2030s.

Labor: the constraint behind the constraint

Manufacturing labor limits the supply response. Much US manufacturing labor has migrated to China and Mexico. Equipment makers in rural locations have land, backlog, and demand, and tell him directly that they cannot get the people. Plants near large metros such as Dallas-Fort Worth, at 8.6m and growing, have better access to labor.

Construction labor is the number two bottleneck. Medium-voltage electricians sit at the top of the scarcity list. The work is mission-critical, and errors destroy expensive equipment or cause serious injury or fatality. Welders, pipefitters, and HVAC technicians are all in high demand, the last increasingly so with liquid cooling. Siting compounds the problem, since Amarillo, El Paso, Fort Stockton, and Pecos combine cheap land and available power with no local labor pool.

The demographics are structural. Around 11,000 Americans retire every week and the trend has further to run. Roughly 34% of construction labor has historically been foreign-born and that inflow has slowed. c80% of high-school students are steered toward college against c5% toward trades.

Pay is the offset. A travelling electrician can earn cUSD250k and skilled mechanical trades cUSD150k, in exchange for mobility, overtime, weekends, and outdoor work on sites of 8,000-10,000 workers. He is explicit about the opportunity. A career started at 18 without college debt is, in his view, a route to rebuilding the middle class that US manufacturing supported in the 1970s and 1980s.

How sites get chosen now

Power feasibility comes first. Available capacity, service voltages, required transmission upgrades, substation requirements, costs, and credible energization dates. A modern facility consumes what a small city does, so the old model of waiting for the utility to build the infrastructure and letting it raise rates has been replaced by hyperscalers funding substations and capacity upgrades themselves, typically alongside behind-the-meter bridging power.

Then everything else. Tenant validation and facility type, whether training, inference, edge, colocation, or enterprise. Access to transmission and natural gas. Water availability against a liquid-cooled design. Hardened fiber. The emissions profile and whether it will be approved, or whether fuel cells or small modular reactors are the better route. He noted that 2032 turbine quotes are themselves pushing developers to evaluate SMRs on the same timeline.

Stranded energy is the differentiator. West Texas flare stacks represent natural gas being burned for want of a pipeline, so siting a facility there converts near-free gas into islanded microgrid power with better emissions control than flaring. Equally, wind farms that cost more to run than their power sells for make excellent sites with faster grid connection. He described the people locating these pockets as the new landmen. ERCOT's proposed 765kV line aims to move power from the energy-dense west to the Houston-Dallas-Austin triangle, where 70% of Texans live.

Natural disaster risk is a live screen. Houston, Galveston, and Corpus Christi carry hurricane exposure, which is why relatively little is built in Florida and why the Austin-San Antonio-Dallas corridor has been favored. The USD50bn Hyperion project in Monroe, Louisiana works because it is far enough inland.

And the offshore alternative is real. If building in the US stays this difficult, he expects the manufacturing precedent to repeat. Canada has land, 40m people, and clean hydroelectric power that it already exports into New York. Mexico has the Laredo corridor, Monterrey two hours from Texas in the same time zone, and an established manufacturing base. Switchgear makers in the El Paso and Ciudad Juárez region estimate a third of their labor crosses the border daily, and

Tijuana offers what California will not permit. Europe is actively luring projects, particularly Scandinavia given the region's hydroelectric power, cool climate, and small populations. The Middle East has capital, though the UAE has seen higher geopolitical risk. Asia is building AI corridors. The next nine largest countries by capacity now roughly equal the US.

Training vs Inference

Training gets the headlines; inference gets the growth. The large training campuses such as Stargate and Abilene are the high-profile projects and are in some ways easier to start, given land availability and permitting in remote locations. But with improving models, use cases are expanding and, on his figure, AI is growing more than 1,000% a year, which is driving a shift toward low-latency inference capacity in and around cities. With increasing enterprise adoption, inference workloads have considerable room for further growth.

The map has broadened. Northern Virginia's Loudoun County remains the anchor, and Mr Dentinger cited an estimate that two-thirds of global email traffic passes through it. Dallas-Fort Worth, Chicago, Columbus, Atlanta, and Phoenix follow, with Reno, Las Vegas, and Kansas City emerging. Single mega-projects now anchor Michigan, Wisconsin, Utah, Wyoming, North Dakota, and Louisiana. The sector is no longer six cities with 800-900MW each. Activity has spread wherever pockets of power exist, with the West Coast and New England the main exceptions.

Pricing, utilities, and the political feedback loop

Input inflation is broad-based. Copper prices are at or near record highs. Aluminium and steel are well above historical levels, with c8-10% of global aluminium transiting the Strait of Hormuz. Freight costs are elevated, with diesel having reached cUSD6.50/gallon, a level at which trucking capacity withdraws. Imported equipment now carries the cost twice over.

Utility bills are the political transmission mechanism. Average US utility bills are up c6-7%, well ahead of general inflation, with New Jersey complaining of c20-22% y-o-y. The blame lands on data centers, renewables, or the IRA infrastructure build, and someone has to pay for it.

Hence take-or-pay and curtailment. Utilities increasingly will not fund capacity without a take-or-pay commitment, and increasingly reserve the right to curtail during peak periods. Agreements are typically negotiated at c20-25% curtailment, with peak expected to be c1% of hours. That reduces grid reliability, so projects need more robust backup generation leading to more hybrid configurations. Take 100MW from the grid quickly to market to hyperscalers against an eventual 500MW requirement, bridge the gap with temporary generation, then either relocate that equipment to the next site or retain it as fast-start backup. Mr Dentinger believes the power solution has become materially more complicated and longer dated than before.

What he watches for signs of a slowdown

- **Capex.** All positive for this year and next.
- **Vacancy.** Approximately 1% nationally and lower in Northern Virginia, so extremely tight.
- **Pre-leasing.** 84% pre-leased before completion now increasingly achievable on land and secured power alone, before construction begins.
- **Neoclouds as the first domino.** If a bubble deflates, it shows up where hyperscalers lease rather than build, and those leases can often be exited more easily than the lessor's description implies.

Corroborating anecdote. *Oracle* has said publicly it cannot supply capacity fast enough and is assembling fragments across the US, Europe, and Asia. At Crusoe's Abilene site, OpenAI and Oracle's decision to stop at 1.2GW rather than take a further c600MW read as a negative headline, but was resolved within a day or two at a higher rate with another hyperscaler, a better outcome for the developer.

Underappreciated technologies

Small modular reactors. Hyperscalers have put hundreds of millions behind the tech, partly because aggressive net-zero commitments are going to be missed as sites run on gas, aeroderivatives, reciprocating engines, and even extended coal. Mr Dentinger's historical frame is instructive. The US built two nuclear plants in 30 years and built the rest in the 1970s and 1980s when government funded it against oil embargoes. All the while, China green-lights 10-11 large plants a year. If turbines are quoted for 2032, an SMR on a similar timeline with no emissions, a small footprint, and no pipeline dependency becomes a serious alternative.

Fuel cells and packaged power. Fuel cells also address the emissions constraint and supply cannot keep up. Customized EPODs and PowerPODs are booked out, which is opening the field to new entrants. Mr Dentinger cited a USD735m order won by Regal Rexnord's Thomson Power Systems, a business with PDU and switchgear knowledge that built and staffed a new plant in Texas, centrally located near DFW labor, despite not having produced these units before.

Simple-to-combined-cycle conversion. Most behind-the-meter builds are simple-cycle. If the power crunch tightens further, adding a small heat recovery steam generator and steam turbine to recover additional output becomes attractive. He named American Supercritical and Ecogen Power Systems as working to make this scalable.

800VDC architecture

Directionally right, ahead of the supply chain. NVIDIA is driving full implementation of next-generation 800VDC power architecture, with pilot projects expected in the early part of next year, tied to the next generation of Vera Rubin chips. The benefit is real. Copper content falls 40-45% and the AC/DC/AC conversion losses go away, which makes the facility materially more efficient.

The constraint is solid-state transformers and DC to DC equipment, where manufacturing has to ramp before the architecture can scale. Mr Dentinger expects an extended pilot and test period, continued construction of conventional facilities, particularly retrofits, and adoption concentrated in new greenfield sites, with an interim 400VDC sidecar route as the likely bridge.

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